

Dr. Hamid Badri

Lead Data Scientist, Battery Modeling & Diagnosis

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Data scientist with a PhD in physics and numerical modeling. I lead battery modeling and diagnosis at Circunomics: state-of-health estimation, degradation and remaining-life modeling for EV and stationary batteries in first and second life. My research background is in inverse problems and adjoint-state methods; I combine physics-based and machine-learning models and take them from data pipelines to production.

EXPERIENCE

Aug 2021 – now

Circunomics GmbH · battery lifecycle analytics & marketplace

Lead Data Scientist, Battery Modeling & Diagnosis (current) · Senior Data Scientist from Aug 2021

- Lead battery state-of-health, degradation and remaining-life modeling for EV and stationary storage batteries, in first and second life.
- Multivariate time-series forecasting, predictive maintenance and anomaly detection with machine learning and deep learning.
- Built and lead the data pipelines in Apache Airflow on Kubernetes (Azure AKS), and the analytics app that serves KPIs, time series and detected anomalies to customers.
- Present the analytics to customers and drive new data-driven product features.

Jan – Jul 2021

Data Scientist · BIAL-R / Merck Group (M-Lab)

- Multivariate time-series forecasting and predictive maintenance with deep learning and machine learning.
- Wrapped models in REST APIs and prepared them for serving on AWS Lambda.

Mar 2019 – Aug 2020 **Data Scientist** · Stockbooking, France

- Stock optimisation with reinforcement learning (Q-learning, SARSA, deep Q-networks).
- Demand forecasting with LSTM and SARIMAX; data pipelines; API serving forecasting and decision models.
- Managed the company's part of ICONET, an EU research project on the Physical Internet.

2015 – 2018

Doctoral Researcher · University of Strasbourg

Inverse problems and model calibration: identification of parameters and source terms for flow and transport in porous and fractured media, with continuous and discrete adjoint states and constrained optimisation. Teaching assistant, Faculty of Physics & Engineering, 2016–2018.

2010 – 2015

Earlier engineering work

- 2015: nonlinear thermal and multi-physics model of Li-ion batteries, ICube / QEERI (Qatar), 5 months.
- 2014: consultant, Qatar Environment and Energy Research Institute.
- 2013: technical manager, GREEN-CAR solar-car prototype (Morocco).
- 2012: graduation project at OCP SA (Morocco): plant availability, automation system and operator GUI.
- 2010–2011: technical manager of a 10-person team, ENSAcAR, Shell Eco-marathon Europe 2011.

SKILLS

BATTERY ANALYTICS

SoH and SoC estimation, degradation and RUL modeling, knee detection, rainflow cycle counting, PyBaMM, second-life assessment.

DATA & MLOPS

Python, pandas, NumPy, Dask, SQL, Apache Airflow, Docker, Kubernetes (AKS), Azure, AWS, Git, Flask, FastAPI.

MACHINE LEARNING

Time-series forecasting, anomaly detection, predictive maintenance, reinforcement learning, SHAP. TensorFlow, Keras, scikit-learn.

MODELING & MATHS

Inverse problems, adjoint-state methods, constrained optimisation, PDE-based simulation, MATLAB, SciPy.

PUBLICATIONS

- 2019 Heterogeneous hydraulic conductivity and porosity fields reconstruction through steady-state flow and transient solute transport data using the continuous adjoint state. F. Delay, H. Badri, P. Ackerer. *Advances in Water Resources*.
- 2018 Identification des paramètres de l'écoulement et du transport en milieu poreux par la méthode de l'état adjoint continu. PhD thesis, University of Strasbourg.
- 2017 A comparison of discrete versus continuous adjoint states to invert groundwater flow in heterogeneous dual porosity systems. F. Delay, H. Badri, M. Fahs, P. Ackerer. *Advances in Water Resources*.
- 2015 Self-consistent approach to solving the 1D Thomas-Fermi equation using an exponential basis set. H. Badri, F. H. Alharbi, R. Jovanovic. *AIP Conference Proceedings*.

Reviewer for Hydrogeology Journal since 2018.

EDUCATION

- 2018 **PhD, Physics and Numerical Modeling** · University of Strasbourg
- 2015 **MSc, Computational Engineering** · University of Strasbourg, ICube
- 2012 **Engineering degree, Process, Energy & Environment** · ENSA Agadir, Morocco
- 2009 **BSc, Science & Technology (Maths, CS, Physics)** · **Energy** · FSTS, Morocco

LANGUAGES & MORE

LANGUAGES

French and Arabic (native), English (fluent, professional).

ROBOTICS

Building a 3D-printed robot arm from scratch; videos on YouTube [@dsc-dev](#).

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